

STRONG MULTIPLICITY ONE FOR SELBERG'S CLASS

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Abstract. Let F and G be two elements in the Selberg class of Dirichlet series. Write $F(s) = \sum_{n=1}^{\infty} a_n(F)/n^s$ and $G(s) = \sum_{n=1}^{\infty} a_n(G)/n^s$. If for all but finitely many primes p , $a_p(F) = a_p(G)$ and $a_{p^2}(F) = a_{p^2}(G)$ then $F = G$. We also show the following. Let $Z_F(T)$ be the multiset of zeroes of $F(s)$ in the region $\text{Re}(s) \geq 1/2$, $|\text{Im}(s)| \leq T$. Suppose that as $T \rightarrow \infty$,

$$|Z_F(T) \Delta Z_G(T)| = o(T),$$

where Δ denotes the symmetric difference. Then $F = G$.

Multiplicité un forte pour la classe de Selberg

Resumé. Soient F et G deux fonctions dans la classe de Selberg. Ecrivons $F(s) = \sum_{n=1}^{\infty} a_n(F)/n^s$ et $G(s) = \sum_{n=1}^{\infty} a_n(G)/n^s$. Si pour presque tous les nombres premiers p , $a_p(F) = a_p(G)$ et $a_{p^2}(F) = a_{p^2}(G)$, alors $F = G$. Nous démontrons aussi le résultat suivant. Soit $Z_F(T)$ l'ensemble des zéros de $F(s)$ dans la région $\text{Re}(s) \geq 1/2$ et $|\text{Im}(s)| \leq T$, comptés avec multiplicité. Nous supposons que lorsque $T \rightarrow \infty$, on a

$$|Z_F(T) \Delta Z_G(T)| = o(T)$$

où Δ désigne la différence symétrique. Alors on a $F = G$.

Version française abrégée La classe de Selberg est l'ensemble $\mathcal{S}$ des fonctions $F(s)$ avec les propriétés suivantes:

- (i) (série de Dirichlet) pour $\text{Re}(s) > 1$, on a l'égalité

$$F(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

où $a_1 = 1$. (Nous écrivons $a_n(F) = a_n$ pour les coefficients de l'écriture de F en série de Dirichlet.);

- (ii) (prolongement analytique) il existe un entier $m \geq 0$, tel que $(s-1)^m F(s)$ ait un prolongement en une fonction entière d'ordre fini;
- (iii) (équation fonctionnelle) il existe des nombres $Q > 0$, $\alpha_i > 0$, et $r_i \in \mathbf{C}$ avec $\text{Re}(r_i) \geq 0$ tels que la fonction

$$\Phi(s) = Q^s \prod_{i=1}^d \Gamma(\alpha_i s + r_i) F(s)$$

vérifie l'équation fonctionnelle

$$\Phi(s) = w \overline{\Phi(1 - \bar{s})}$$

¹ Research partially supported by NSERC and FCAR grants.

² Research partially supported by NSERC.

- où w est une nombre complexe avec $|w| = 1$.
(iv) (produit eulérien) la fonction $F(s)$ s'écrit

$$F(s) = \prod_p F_p(s)$$

où

$$F_p(s) = \exp \left(\sum_{k=1}^{\infty} \frac{b_{p^k}}{p^{ks}} \right)$$

où $b_{p^k} = O(p^{k\theta})$ pour un certain $\theta < 1/2$ et p désigne une nombre premier. (Nous écrivons $b_p(F) = b_p$.)

- (v) (l'hypothèse de Ramanujan) pour tout $\epsilon > 0$ fixé, on a l'égalité

$$a_n = O(n^\epsilon).$$

Une fonction $F \in \mathcal{S}$ est dite **primitive** s'il est impossible d'écrire $F = F_1 F_2$ avec $F_1, F_2 \in \mathcal{S}, F_1 \neq 1, F_2 \neq 1$. Selberg a conjecturé

- (a) pour toute fonction primitive F , on a l'égalité

$$\sum_{p \leq x} \frac{|a_p(F)|^2}{p} = \log \log x + O(1);$$

- (b) si F et G sont des fonctions primitives distinctes, on a l'egalité

$$\sum_{p \leq x} \frac{a_p(F) \overline{a_p(G)}}{p} = O(1)$$

quand $x \rightarrow \infty$.

Nous savons que chaque élément de $\mathcal{S}$ a une factorisation en produit de fonctions primitives. Ces conjectures impliquent que cette factorisation est unique. Elles impliquent aussi que, si pour presque tout p , on a l'égalité $a_p(F) = a_p(G)$, alors on a $F = G$. (Une propriété est vraie, par définition, **pour presque tout** p , si le nombre d'exceptions en p est fini.)

Théorème 1. Supposons que les égalités $a_p(F) = a_p(G), a_{p^2}(F) = a_{p^2}(G)$ soient vraies pour presque tout p . Alors on a $F = G$.

Corollaire. Si l'égalité $F_p(s) = G_p(s)$ est vraie pour presque tout p , alors on a $F = G$.

Théorème 2. Soit $F \in \mathcal{S}$, et soit $Z_F(T)$ l'ensemble des zéros de $F(s)$, comptés avec multiplicité, dans la région $\text{Re}(s) \geq 1/2, |\text{Im}(s)| \leq T$. Pour $F, G \in \mathcal{S}$ supposons que la différence symétrique vérifie

$$|Z_F(T) \Delta Z_G(T)| = o(T)$$

quand $T \rightarrow \infty$. Alors on a $F = G$.

La preuve du théorème 1 dépend du lemme suivant:

Lemme. Soit $f(s)$ une fonction méromorphe sur $\mathbf{C}$, sans pôle dans le demi-plan fermé $\operatorname{Re}(s) \geq 1/2$, et ne s'annulant pas dans ce demi plan. Supposons que $\log f(s)$ est une série de Dirichlet et qu'il existe $A, \alpha_i, \gamma_i > 0, \beta_i, \delta_i \in \mathbf{C}$ avec $\operatorname{Re}(\beta_i), \operatorname{Re}(\delta_i) \geq 0$ tels que la fonction

$$H(s) = A^s \frac{\prod_{i=1}^{d_1} \Gamma(\alpha_i s + \beta_i)}{\prod_{i=1}^{d_2} \Gamma(\gamma_i s + \delta_i)} f(s)$$

verifie l'equation fonctionnelle

$$H(s) = w \overline{H(1 - \bar{s})}$$

pour un certain nombre complexe w de module 1. Alors $f(s)$ est une constante.

La preuve du théorème 2 commence par une généralisation d'une formule classique de Landau:

$$-b_n(F) \log n = 2\pi \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{0 < \operatorname{Im}(\rho) < T} n^\rho$$

où la somme est faite sur les zéros ρ de $F(s)$.

The Selberg class $\mathcal{S}$ consists of functions $F(s)$ of a complex variable s satisfying the following properties:

(i) (Dirichlet series) for $\operatorname{Re}(s) > 1$,

$$F(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

where $a_1 = 1$. (We will write $a_n(F) = a_n$ for the coefficients of the Dirichlet series of F .);

(ii) (Analytic continuation) for some integer $m \geq 0$, $(s-1)^m F(s)$ extends to an entire function of finite order;

(iii) (Functional equation) there are numbers $Q > 0, \alpha_i > 0$ and $r_i \in \mathbf{C}$ with $\operatorname{Re}(r_i) \geq 0$ so that

$$\Phi(s) = Q^s \prod_{i=1}^d \Gamma(\alpha_i s + r_i) F(s)$$

satisfies

$$\Phi(s) = w \overline{\Phi(1 - \bar{s})}$$

where w is a complex number with $|w| = 1$ and $\overline{\Phi}(s) = \overline{\Phi(\bar{s})}$;

(iv) (Euler product)

$$F(s) = \prod_p F_p(s)$$

where

$$F_p(s) = \exp \left(\sum_{k=1}^{\infty} \frac{b_{p^k}}{p^{ks}} \right)$$

where $b_{p^k} = O(p^{k\theta})$ for some $\theta < 1/2$ and p denotes a prime number here and throughout the paper. (We shall write $b_p(F) = b_p$.)

(v) (Ramanujan hypothesis) For any fixed $\epsilon > 0$, $a_n = O(n^\epsilon)$.

A function $F \in \mathcal{S}$ is called **primitive** if it cannot be written as a non-trivial product of two elements in $\mathcal{S}$. It can be shown that every element in $\mathcal{S}$ can be written as a product of primitive elements. (See [1] and [2].) Selberg [4, (1.12) and (1.13)] conjectures:

(a) for any primitive function F ,

$$\sum_{p \leq x} \frac{|a_p(F)|^2}{p} = \log \log x + O(1);$$

(b) for two distinct primitive functions F and G ,

$$\sum_{p \leq x} \frac{a_p(F) \overline{a_p(G)}}{p} = O(1).$$

These conjectures imply that the factorization into primitive elements is unique. It is easy to see that the conjectures also imply that if $a_p(F) = a_p(G)$ for all but finitely many primes p , then $F = G$. This looks difficult to prove unconditionally. We will show:

Theorem 1. Suppose $a_p(F) = a_p(G)$, $a_{p^2}(F) = a_{p^2}(G)$ for all but finitely many primes p . Then $F = G$.

Corollary . If $F_p(s) = G_p(s)$ for all but finitely many primes p , then $F = G$.

The corollary can be viewed as an analogue of the strong multiplicity one theorem for automorphic L -functions on GL_n .

One can also prove analogously:

Theorem 2. For $F \in \mathcal{S}$, denote by $Z_F(T)$ the multiset of zeroes of $F(s)$ in the region $\text{Re}(s) \geq 1/2$, $|\text{Im}(s)| \leq T$. For $F, G \in \mathcal{S}$, suppose the symmetric difference satisfies

$$|Z_F(T) \Delta Z_G(T)| = o(T)$$

as $T \rightarrow \infty$. Then $F = G$.

We begin by noting that in view of the functional equation and the Phragmén-Lindelöf theorem, any entire element of $\mathcal{S}$ is of order 1. (In general, if $F(s)$ has a pole of order m at $s = 1$, then $(s - 1)^m F(s)$ is necessarily of order 1.) The proof of Theorem 1 depends on the following lemma.

Lemma. Let $f(s)$ be a meromorphic function on $\mathbf{C}$, analytic for $\text{Re}(s) \geq 1/2$ and non-vanishing there. Suppose that $\log f(s)$ is a Dirichlet series and that $f(s)$ satisfies the functional equation

$$H(s) = w \overline{H}(1 - s)$$

where w is some complex number of absolute value 1, and

$$H(s) = A^s \frac{\prod_{i=1}^{d_1} \Gamma(\alpha_i s + \beta_i)}{\prod_{i=1}^{d_2} \Gamma(\gamma_i s + \delta_i)} f(s)$$

with certain $A, \alpha_i, \gamma_i > 0$ and $\operatorname{Re}(\beta_i), \operatorname{Re}(\delta_i) \geq 0$. Then $f(s)$ is constant.

Proof. Recall that [5, p. 313] for any Dirichlet series $F(s)$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T F(\sigma + it) y^{\sigma + it} dt = \begin{cases} a_n(F) & \text{if } n = y \\ 0 & \text{otherwise} \end{cases}$$

where $\sigma > \sigma_a(F)$ the abscissa of absolute convergence of F . In other words,

$$\frac{1}{2\pi i} \int_{(\sigma)} F(s) y^s ds = \infty$$

if $y = n$ and $a_n(F) \neq 0$. From the functional equation and the fact that the factor corresponding to the gamma functions introduce poles only in $\operatorname{Re}(s) \leq 0$, we find that $H(s)$ is analytic in $\operatorname{Re}(s) \geq 1/2$ and does not vanish in this half-plane. By the functional equation, the same is true in $\operatorname{Re}(s) \leq 1/2$. Thus $H(s)$ is entire. Moreover, Stirling's formula and the functional equation imply that it is of order 1. By Hadamard's theorem, it follows that $H(s) = e^{as+b}$ for some constants a and b . Hence,

$$\frac{f'}{f}(s) = a - \log A + \sum_{i=1}^{d_2} \frac{\Gamma'}{\Gamma}(\gamma_i s + \delta_i) \gamma_i - \sum_{i=1}^{d_1} \frac{\Gamma'}{\Gamma}(\alpha_i s + \beta_i) \alpha_i$$

is a Dirichlet series. The derivative of this is again a Dirichlet series. Since

$$\frac{d}{ds} \left(\frac{\Gamma'}{\Gamma}(s) \right) = - \sum_{m=0}^{\infty} \frac{1}{(s+m)^2}$$

and since

$$\frac{1}{2\pi i} \int_{(\sigma)} \frac{y^s}{(\alpha s + \beta)^2} ds = \begin{cases} \alpha^{-2} y^{-\beta/\alpha} \log y & \text{if } y > 1 \\ 0 & \text{if } y \leq 1 \end{cases}$$

for σ sufficiently large, we deduce by an easy computation that

$$\frac{1}{2\pi i} \int_{(\sigma)} \frac{d}{ds} \left(\frac{f'}{f}(s) \right) y^s dy = O(1)$$

for any $y \geq 1$. Hence, every coefficient of $d/ds(f'/f(s))$ is zero. As f'/f is a Dirichlet series, this implies that $f'/f = 0$. Hence f is constant. This completes the proof of the lemma.

We can now prove Theorem 1. Set

$$f(s) = \prod_p (F_p(s)/G_p(s)).$$

Since $\log F_p(s)$ is an absolutely convergent Dirichlet series for $\operatorname{Re}(s) > \theta$, we deduce that $F_p(s)$ is absolutely convergent for $\operatorname{Re}(s) > \theta$ and is non-vanishing there. In particular, this holds for $\operatorname{Re}(s) \geq 1/2$. We can factor

$$\left(1 + \frac{a_p(F)}{p^s} + \frac{a_{p^2}(F)}{p^{2s}}\right)^{-1}$$

from the numerator and denominator of $F_p(s)/G_p(s)$ and write

$$f(s) = \prod_p f_p(s)$$

where each $f_p(s)$ converges absolutely for $\operatorname{Re}(s) > \theta$ (for some $\theta < 1/2$, and does not vanish there. $f(s)$ satisfies the conditions of the lemma. Hence $f(s)$ is constant which must be 1 because that is the constant term of the Dirichlet series. Therefore $F = G$. This completes the proof of Theorem 1.

Set $b_x(F) = 0$ unless x is a prime power. To prove Theorem 2, we begin by noting the formula

$$-b_n(F) \log n = 2\pi \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{0 < \operatorname{Im}(\rho) < T} n^\rho$$

where the sum is over the zeroes ρ of $F(s)$. This is the analogue of a classical formula of Landau for the Riemann zeta function. It is easily derived from the integral identity used in the beginning of the proof of the lemma and by the method of [6, pp.61-62] in conjunction with the lemmas of [3, p.540-544]. Indeed, let q be a positive number, less than the ordinate of any zero of F in the region $\operatorname{Re}(s) > 0$ and $\operatorname{Im}(s) > 0$. Moreover, we suppose that $q > \max_{1 \leq i \leq d} |\operatorname{Im}(r_i)/\alpha_i|$ where r_i and α_i are as in (iii) in the definition of the Selberg class. Then,

$$\left| \int_{2+iq}^{2+iT} \frac{F'}{F}(s) x^s ds + i(T-q)b_x(F) \log x \right| \leq 2x^2 \sum_{\substack{n=1 \\ n \neq x}}^{\infty} \frac{|b_n(F)| \log n}{n^2 |\log(x/n)|}.$$

By Cauchy's theorem,

$$\int_{2+iq}^{2+iT} \frac{F'}{F}(s) x^s ds - 2\pi i \sum_{q < \operatorname{Im}(\rho) < T} x^\rho = \int_{2+iq}^{a+iq} + \int_{a+iq}^{a+iT} + \int_{a+iT}^{2+iT} = I_1 + I_2 + I_3 \quad (\text{say})$$

where $a < 0$ and T is not the ordinate of any zero. It is also seen that by the analogue of lemma 2 of [3, p.540],

$$\frac{F'}{F}(s) = O(\log |s|)$$

for $\sigma \leq -1, t \geq q$. Thus, the integral I_2 tends to zero as $a \rightarrow -\infty$ and so

$$\int_{2+iq}^{2+iT} \frac{F'}{F}(s) x^s ds - 2\pi i \sum_{q < \text{Im}(\rho) < T} x^\rho = \int_{2+iq}^{-\infty+iq} + \int_{-\infty+iT}^{2+iT}.$$

Again, we find

$$\int_{-\infty+iT}^{-1+iT} \frac{F'}{F}(s) x^s ds = \int_{-\infty}^{-1} O(\log(|\sigma| + T)) x^\sigma d\sigma = O(\log T)$$

which applies for $T = q$ also. Now by the analogue of lemma 5 of [3,p.542] we find

$$\int_{-1+iT}^{2+iT} \frac{F'}{F}(s) x^s ds = \sum_{|T-\gamma| < 1} \int_{-1+iT}^{2+iT} \frac{x^s}{s-\rho} ds + O(\log T).$$

We handle the integral in the sum as in [6, p. 62]: write

$$\int_{-1+iT}^{2+iT} = \int_{-1+iT}^{-1+i(T+2)} + \int_{-1+i(T+2)}^{2+i(T+2)} + \int_{2+i(T+2)}^{2+iT}$$

and notice that each term is $O(1)$ and there are $O(\log T)$ terms so we deduce that the sum on the right is $O(\log T)$. Since there are only a finite number of zeroes ρ with $0 < \text{Im}(\rho) \leq q$, we deduce that

$$\frac{-b_x(F) \log x}{2\pi} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{0 < \text{Im}(\rho) < T} x^\rho.$$

To prove the theorem, we have by hypothesis,

$$\begin{aligned} \frac{-b_x(F) \log x}{2\pi} &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{\substack{0 < \text{Im}(\rho) < T \\ F(\rho)=0}} x^\rho \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\sum_{\substack{0 < \text{Im}(\rho) < T \\ G(\rho)=0}} x^\rho + o(T) \right) \\ &= \frac{-b_x(G) \log x}{2\pi} \end{aligned}$$

so that $F = G$. This completes the proof.

It is clear that the proof actually establishes a stronger result. Let I_T be an interval contained in $[0, T]$ and let $Z_F(I_T)$ denote the multiset of zeroes of F whose imaginary parts are contained in I_T . Suppose that $|I_T| = f(T) \log T$ with $f(T) \rightarrow \infty$ as $T \rightarrow \infty$. If $|Z_F(I_T) \Delta Z_G(I_T)| = o(|I_T|)$ as $T \rightarrow \infty$, then $F = G$.

REFERENCES

- [1] B. Conrey and A. Ghosh, On the Selberg class of Dirichlet series: Small weights, preprint.
- [2] M. Ram Murty, Selberg's conjectures and Artin L -functions, *Bulletin of the Amer. Math. Soc.*, (1994) to appear.
- [3] V. Kumar Murty, Explicit formulae and the Lang-Trotter conjecture, *Rocky Mountain Journal of Mathematics*, **15** (1985) 535-551.
- [4] A. Selberg, Old and new conjectures and results about a class of Dirichlet series, Collected Papers, Vol. 2, pp. 47-63, Springer-Verlag, 1991.
- [5] E.C. Titchmarsh, The Theory of Functions, Oxford University Press, Second Edition, 1939.
- [6] E.C. Titchmarsh, The zeta function of Riemann, Cambridge Tracts in Mathematics, No. 26, Cambridge University Press, 1930.

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